

Pharmacy Inventory Data Analysis Project

A data analytics project on pharmacy inventory management — covering data cleaning, exploratory analysis, SQL querying, and inventory optimization (EOQ, Safety Stock, Reorder Point, Stockout Rate).

📁 Files in This Repo

- `pharmacy_inventory_data_raw.csv` — raw dataset
 - `pharmacy_inventory_data_clean.xlsx` — cleaned dataset
 - `EDA_using_python.ipynb` — exploratory analysis (Python)
 - `sql_queries.ipynb` — SQL-based analysis
 - `basic_analysis_reports.xlsx` / `businessanalyst_reports.xlsx` — final reports
 - `dashboard_report.pdf` — Power BI dashboard exported as PDF (visualization)
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🔧 Steps I Took

1. Created a synthetic pharmacy inventory dataset — 500 medicines across 12 months (Aug 2025-Jul 2026), with stock, demand, cost, and lead time data.
 2. Cleaned and validated the raw data (`pharmacy_inventory_data_clean.xlsx`).
 3. Performed EDA in Python — demand trends, stockout patterns, category/manufacturer breakdowns.
 4. Ran SQL queries for business-level aggregates (monthly demand, stockouts, category summaries).
 5. Built inventory optimization metrics:
 - **EOQ** (Economic Order Quantity)
 - **Safety Stock**
 - **Reorder Point**
 - **Stockout Rate**
 - **EOQ vs Current Order Quantity** cost comparison
 6. Compiled findings into Excel reports.
 7. Built a Power BI dashboard to visualize key metrics (stock levels, demand trends, stockout rates, EOQ savings) and exported it as a PDF, included in the report file for easy viewing on GitHub.
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🏢 Business Problems This Solves

1. **Over/under-ordering** → EOQ gives the cost-optimal order quantity per medicine.
2. **Stockouts of critical medicines** → Stockout Rate flags medicines with the highest unmet demand.
3. **Demand uncertainty** → Safety Stock provides a buffer based on demand variability.
4. **Reactive reordering** → Reorder Point gives a clear stock threshold to trigger new orders.
5. **Cost inefficiency** → EOQ vs Current comparison quantifies savings from switching to EOQ-based ordering.

Goal: Reduce total inventory cost while minimizing stockouts, by moving from intuition-based to data-driven ordering.

⚠️ Key Assumptions

- Dataset is **artificially/synthetically generated** — not real pharmacy data.
- 500 medicines across 12 categories assumed to represent a mid-to-large pharmacy.
- Demand per medicine per month is independent (no substitution effects modeled).
- Lead time, unit cost, and holding/ordering costs assumed constant per medicine over the period.
- Stockout units = lost demand (no backordering assumed).
- “Is_Critical” tagging based on general category judgment, not an official clinical source.

- No external disruptions (recalls, outbreaks, supply shocks) modeled.
- All costs in INR.

Tools Used

Python (pandas, matplotlib, seaborn) · SQL · Excel · Power BI · Jupyter Notebook

This is a portfolio project using synthetic data — not intended for real business decisions without validation.